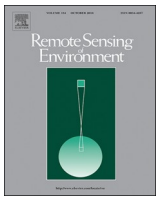
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Robust Landsat-based crop time series modelling



D.P. Roy*, L. Yan

Geospatial Sciences Center of Excellence, 1021 Medary Ave, Wecota Hall, Box 506B, South Dakota State University, Brookings, SD 57007, USA

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ABSTRACT

Reliable satellite monitoring of agriculture is often difficult because surface variations occur rapidly compared to the cloud-free satellite observation frequency. Harmonic time series models, i.e., superimposed sequences of sines and cosines, have an established provenance for fitting satellite vegetation index time series to coarse resolution satellite data, but their application to medium resolution Landsat data for crop monitoring has been limited. Non-linear harmonic models have been shown to perform well over agricultural sites using single-year Moderate Resolution Imaging Spectroradiometer (MODIS) time series, but have not been explored with Landsat data. The 2017 availability of Landsat Analysis Ready Data (ARD) over the United States provides the opportunity to investigate the utility of temporally rich Landsat data for 30 m pixel-level crop monitoring. In this paper, the capability of 5- and 7-parameter linear harmonic models and a 5-parameter non-linear harmonic model applied to a year of Landsat 5 Thematic Mapper (TM) and Landsat 7 Enhanced Thematic Mapper Plus (ETM+) ARD is investigated. The analysis is undertaken over six sites, each defined by a 5000 × 5000 30 m pixel ARD tile, that together include the major conterminous United States (CONUS) crops identified by inspection of the United States Department of Agriculture (USDA) Cropland Data Layer (CDL). The model fits are evaluated as the root mean square difference (RMSD) between the fitted and the observed Landsat data. Considering locations with at least 21 annual Landsat observations, the 7-parameter linear harmonic model (tile mean crop NDVI RMSD values ranging from 0.052 to 0.072) and the 5-parameter non-linear harmonic model (tile mean crop NDVI RMSD values ranging from 0.054 to 0.074) are shown to be able to fit annual Landsat NDVI time series for most CONUS crops, whereas the 5-parameter linear harmonic model cannot (tile mean crop NDVI RMSD values ranging from 0.072 to 0.099). If there are between 15 and 20 annual Landsat observations, the 5-parameter non-linear harmonic model is recommended for fitting annual NDVI crop time series, and if there are ≥ 21 observations, then either the 5-parameter non-linear or the 7-parameter linear model can be used. The 7-parameter model had marginally smaller mean NDVI RMSD values but larger standard deviations than the 5-parameter non-linear model, likely due to the relative robustness of the non-linear model to over-fitting and oscillations. None of the models could reliably fit crops with multiple stages, such as alfalfa, that are insufficiently sampled using combined Landsat 5 TM and Landsat 7 ETM+ time series. Given the utility of the growing season peak NDVI for crop yield applications, the date and magnitude of the model fitted peak NDVI are compared to quantify model reporting differences. The differences between the 7-parameter linear and the 5-parameter non-linear harmonic models are not large. For each ARD tile, the mean absolute differences in the estimated peak NDVI days varied from < 2 days in the northern ARD tiles, which had short growing seasons and similar crops, to less than a week for the other tiles except for nearly 10 days for the California tile that had longer growing seasons and more diverse crops including crops with multiple stages. The paper concludes with a discussion and recommendations for future research.

1. Introduction

There is an outstanding expectation for systematic large area land surface monitoring using medium resolution (10 m^2 to 100 m^2 pixel) satellite data. Landsat data have been used for agricultural applications since the first Landsat sensors data became available (Bauer et al., 1978), but the majority of large area Landsat monitoring algorithms have focused on mapping land cover and change (Wulder et al., 2018). The Landsat temporal resolution is low; for example, the combination of two contemporaneous satellites nominally provides only an eight-day revisit (Loveland and Dwyer, 2012), which is often reduced by cloud cover (Kovalskyy and Roy, 2013). Reliable temporal monitoring

* Corresponding author.
E-mail address: david.roy@sdstate.edu (D.P. Roy).

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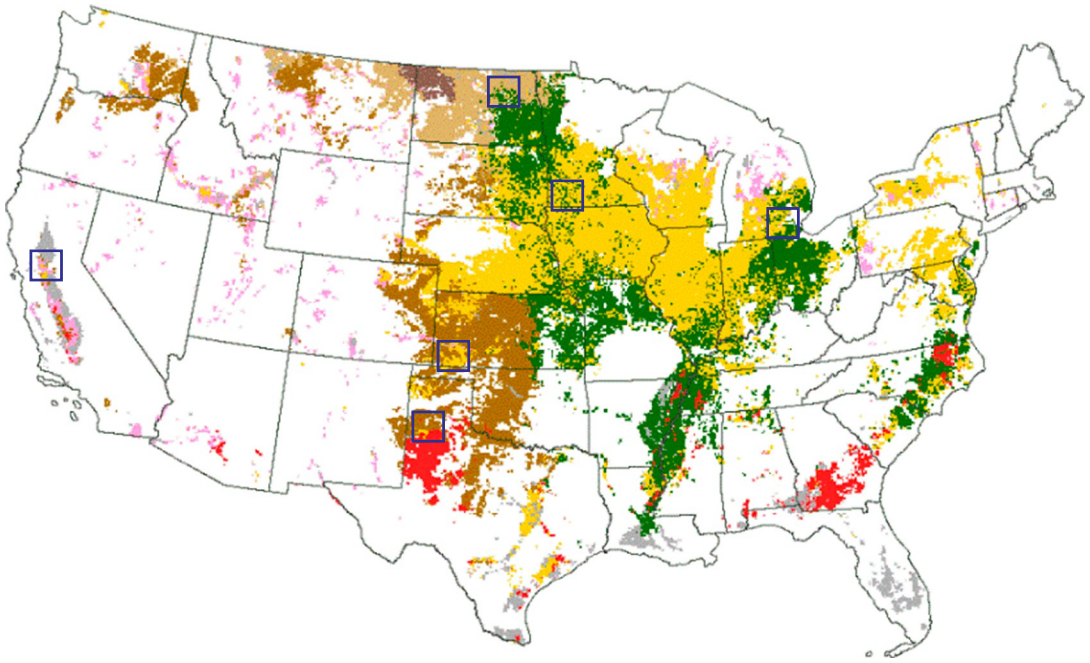


Fig. 1. Six Landsat 150 × 150 km ARD tiles (blue squares) selected to capture different major agricultural crop types in the CONUS. The background image shows a CONUS 2010 CDL majority crop map (from Yan and Roy, 2016). The majority CDL classes in 7.5 km × 7.5 km gridcells with ≥10% crop pixels are shown with the original CDL color legend for the major crops of corn, soybeans, alfalfa, winter wheat, spring wheat, durum wheat, and cotton; the other crops are shown in gray. Gridcells with < 10% crop pixels are shown in white and state boundaries are shown in black. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

is difficult if surface variations occur rapidly relative to the cloud free satellite observation frequency. This is often true for arable agriculture; for example, double or multiple cropped agricultural systems such as alfalfa, or winter wheat, may require near-daily observations to capture crop variability (Wardlow et al., 2007; Gao et al., 2017; Houborg and McCabe, 2018). Approaches that combine several years of Landsat data together to derive a representative annual product, or to compare new observations against, may not work if there is inter-annual variability in the surface cover and/or condition. In agricultural regions, this is often the case due to crop rotation and fallowing. For example, biennial rotation between corn and soybeans is common in the United States (Plourde et al., 2013) and these crops have different phenology and mature stage spectral signatures (Yan and Roy, 2015; Gao et al., 2017). Despite these challenges, there is an established history of large area agricultural remote sensing that incorporate or solely use Landsat satellite data, including agricultural crop type classification (Johnson, 2013; Song et al., 2017; Cai et al., 2018), crop non-crop mapping (Xiong et al., 2017), field extraction (Yan and Roy, 2016; Graesser and Ramankutty, 2017), crop status assessment (Lenney et al., 1996; Gitelson et al., 2012; Simic Milas and Vincent, 2017), and yield prediction (Tennakoon et al., 1992; Azzari et al., 2017).

The recent 2017 availability of Landsat Analysis Ready Data (ARD) (USGS, 2018) has significantly reduced the complexity of Landsat time series data handling and pre-processing (Egorov et al., 2018). Combination of two contemporaneous Landsat satellites over the United States, where every Landsat acquisition is obtained (Wulder et al., 2016), provides the opportunity to investigate the utility of temporally rich Landsat data for pixel level crop monitoring. In particular, there is considerable research on the use of satellite time series for crop status and yield estimation and forecasting. This is likely due to the commercial value of these information, combined with free and open satellite availability, and increasing computational processing capabilities.

Model fitting of satellite time series has an established provenance developed using harmonic models, i.e., superimposed sequences of sines and cosines, fitted to vegetation index time series from coarse spatial but high temporal resolution Advanced Very-High-Resolution Radiometer (AVHRR) and Moderate Resolution Imaging Spectroradiometer (MODIS) data (Menenti et al., 1993, Olsson and Eklundh, 1994, Moody and Johnson, 2001, Jakubauskas et al., 2001, Hermance, 2007, Scharlemann et al., 2008). Time series model fitting approaches have more recently been applied to Landsat time series. Notably, logistic regression curves have been applied to Landsat vegetation index time series for phenological studies (Fisher et al., 2006; Kovalskyy et al., 2012; Melaas et al., 2013) and linear harmonic models have been applied to fill missing Landsat observations and to predict reflectance at any date for land cover change applications (Brooks et al., 2012; Zhu et al., 2015; Chen et al., 2017; Pouliot and Latifovic, 2018; Yan and Roy, 2018). Non-linear harmonic models have been shown to perform well over agricultural sites using MODIS time series (Carrão et al., 2010) but have not been explored using Landsat time series.

The objective of this paper is to compare the capability of linear and non-linear harmonic models to fit time series models to Landsat vegetation index time series over predominantly agricultural crop sites and using only a single year of data. A year of Landsat 5 Thematic Mapper (TM) and Landsat 7 Enhanced Thematic Mapper Plus (ETM+) ARD for six agricultural sites that include the major crops in the conterminous United States (CONUS) are considered. Time series of normalized vegetation difference index (NDVI), which are atmospherically corrected and filtered to remove clouds, shadow, snow and saturated pixels, are used to fit 5- and 7-parameter linear harmonic models and to fit a non-linear 5-parameter harmonic model. The annual model fits are evaluated as the root mean square difference between the model-predicted and the observed Landsat data. Further, the differences between the model-predicted day of year and magnitude of growing season NDVI peaks are quantified. The results are discussed in terms of the applicability of the models for Landsat-based crop monitoring with recommendations for future research.

Table 1
Crop and non-crop land cover classes in the six ARD tiles derived from the 2010 CDL data (Fig. 3). Only the CDL classes covering > 2% in each tile are summarized.

	North Dakota	Minnesota	Michigan	Kansas	Texas	California
Crop CDL class	Spring Wheat (20.3%), Soybeans (13.5%), Canola (5.2%), Corn (3.2%), Dry Beans (2.9%)	Corn (41.5%), Soybeans (36.1%)	Soybeans (17.6%), Corn (16.4%), Winter Wheat (4.1%), Alfalfa (3.9%)	Winter Wheat (16.4%), Corn (10.5%), Sorghum (5.1%)	Winter Wheat (19.0%), Cotton (17.3%), Corn (5.0%)	Rice (6.7%), Alfalfa (3.1%), Winter Wheat (2.6%), Corn (2.5%), Walnuts (2.4%)
Non-crop CDL class	Grassland/Pasture (29.0%), Herbaceous Wetlands (8.1%), Open Water (6.2%), Developed/Open Space (3.9%)	Grassland/Pasture (8.5%), Developed/Open Space (4.8%), Herbaceous Wetlands (2.7%)	Deciduous Forest (13.8%), Grassland/Pasture (11.0%), Woody Wetlands (10.9%), Developed/Open Space (8.1%), Developed/Low Intensity (6.1%), Developed/Medium Intensity (2.2%)	Grassland/Pasture (50.5%), Fallow/Idle Cropland (9.8%), Developed/Open Space (3.2%)	Grassland/Pasture (42.9%), Shrub/Scrub (7.6%), Fallow/Idle Cropland (2.6%)	Grassland/Pasture (31.0%), Shrub/Scrub (9.2%), Evergreen Forest (8.1%), Fallow/Idle Cropland (5.5%), Developed, Open Space (4.6%), Open Water (3.0%), Developed, Medium Intensity (2.9%), Developed, Low Intensity (2.7%), Herbaceous Wetlands (2.0%)

2. Study areas and data

Landsat 5 TM and Landsat 7 ETM+ ARD were used in this study. The ARD are generated by the United States Geological Survey (USGS, 2018) in non-spatially overlapping, fixed 5000 × 5000 30 m pixel (150 × 150 km) tiles in the Albers Equal Area Conic projection with the same spatial tile structure used to generate the CONUS Web-Enabled Landsat Data (WELD) products (Roy et al., 2010). Each individual orbit of Landsat data is stored independently. Any pixel observations labelled in the ARD as saturated in the red (center wavelength for both sensors ~0.66 μm) or near infrared (NIR) (~0.83 μm) bands, or as cloudy, snow, or shadow were discarded. The atmospherically corrected ARD were used in this study. The ARD Landsat 5 TM and Landsat 7 ETM+ ARD reflectance bands are atmospherically corrected using the established Landsat Ecosystem Disturbance Adaptive Processing System (LEDAPS) Surface Reflectance Algorithm (Masek et al., 2006), which over the CONUS has reported mean reflectance normalized residuals of < 12% (blue band), 6% (green and red bands), 5% (NIR band), and residuals of 1% to 2% for the short wave infrared bands (Ju et al., 2012). The NDVI was derived (as the NIR minus the red surface reflectance divided by their sum) and observations with NDVI < 0 were discarded in an attempt to remove unvegetated ones as they are unimportant for the annual time series fitting described in this paper. Across the CONUS, atmospherically-corrected Landsat NDVI is on average 0.1 greater than top-of-atmosphere NDVI for vegetated surfaces (Roy et al., 2014a), and so the NDVI < 0 threshold is quite conservative.

Landsat 5 TM and Landsat 7 ETM+ ARD for year 2010 were considered. This was because in 2010, the United States Department of Agriculture (USDA) Cropland Data Layer (CDL) (Johnson and Mueller, 2010) was available for all of the CONUS, and also the Landsat 5 and 7 orbit overpass times were similar, unlike in certain previous years when the Landsat 5 orbit was not maintained (Zhang and Roy, 2016). The USDA CDL product was used to locate six ARD tiles that are representative of the major CONUS agricultural crops and that include several crop types (Fig. 1, Table 1). The CDL is well suited for this purpose (Lark et al., 2017) and is generated in the Albers projection at 30 m resolution using a variety of moderate resolution satellite imagery and extensive agricultural ground truth data to define > 100 land cover and crop type classes (Johnson and Mueller, 2010).

The temporal availability of Landsat observations at each ARD tile pixel will influence the per-pixel harmonic model fitting capability. The six tiles (North Dakota, Minnesota, Michigan, Kansas, Texas, and California) have a variable availability of valid Landsat 5 TM and Landsat 7 ETM+ data in 2010. This is illustrated in Fig. 2 that shows the spatial percentage of the pixels in each tile with valid observations for each day. There is considerable daily variation reflecting both temporal variations in cloud, snow and shadow, but also the variable spatial coverage of the Landsat 5 TM and Landsat 7 ETM+ swaths over each tile. The northern tiles (North Dakota, Minnesota and Michigan) have far fewer valid observations in the winter months due primarily to snow cover. Importantly, however, for all the tiles, there is no particular pattern of data availability in the spring to autumn crop growing season.

The 2010 CDL data for each of the six tiles are illustrated in Fig. 3 and the main CDL classes (> 2% of each tile) are summarized in Table 1. The North Dakota tile crops are predominantly spring wheat (20.3% of the 2010 CDL tile pixels) and soybean (13.5%) with a pronounced but short growing season that makes crop classification challenging if there are missing and cloudy satellite data (Chang et al., 2007). Nearly one third (29.0%) of the North Dakota tile is grassland/pasture. The Minnesota tile is predominantly corn (41.5%) and soybean (36.1%), and the predominant non-crop class is also grassland/pasture (8.5%). The Michigan tile has soybean (17.6%) and corn (16.4%) and was selected for this study as it has a large number of non-crop classes and relatively high degree of cloud obscuration (Kovalskyy and Roy,

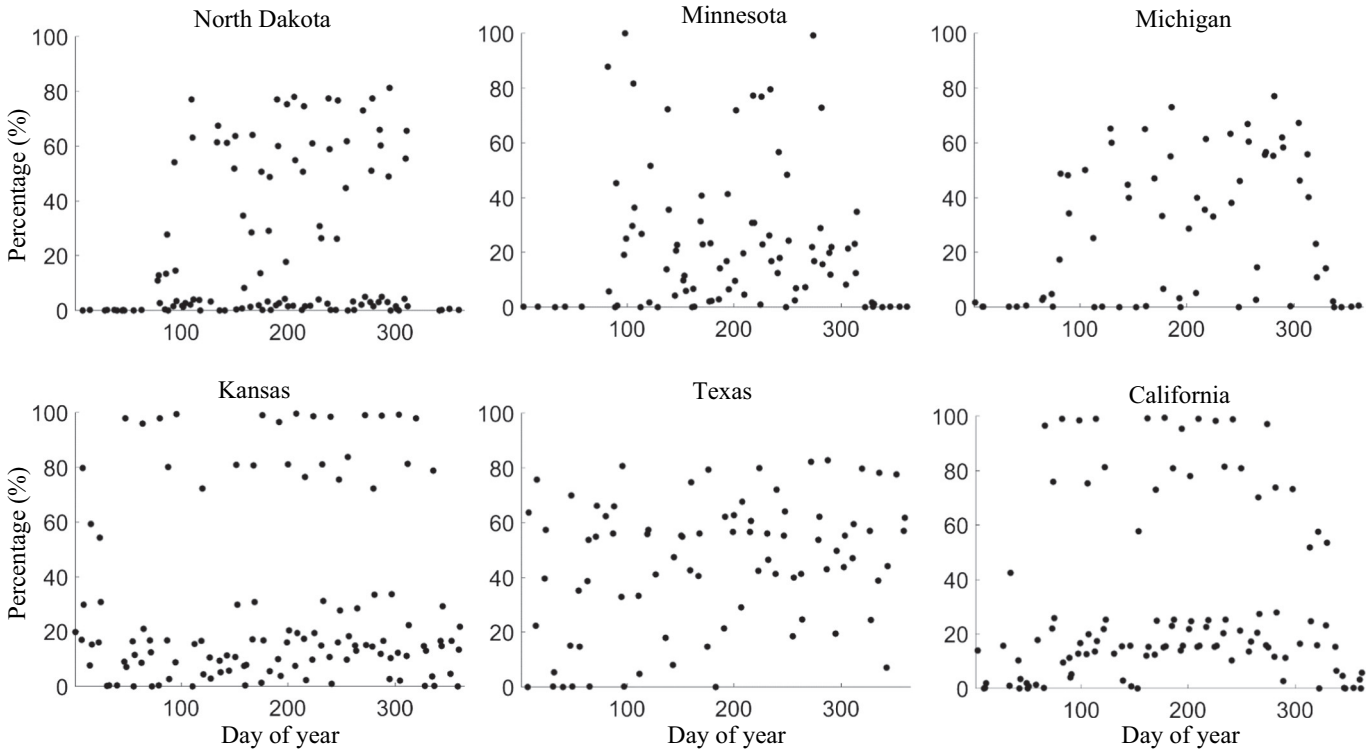


Fig. 2. The spatial percentage of the ARD tile pixels with valid (cloud-free, snow-free, shadow-free, unsaturated, atmospherically corrected NDVI > 0.0) Landsat 5 TM and Landsat 7 ETM+ ARD data in 2010 for the six selected tiles (Fig. 1). The spatial percentages summarize the percentage of ARD tile pixels (from a total of 5000^2 pixels) on each day where there were tile ARD data.

2013) that is evident in Fig. 2. The Kansas tile crops are predominantly winter wheat (16.4%) and corn (10.5%) and the predominant non-crop classes are grassland/pasture (50.5%) and fallow/idle cropland (9.8%). The Texas tile crops are mainly winter wheat (19.0%) and cotton (17.3%), and the non-crop classes are mainly grassland/pasture (42.9%) and shrub/scrub (7.6%). The California tile has the most CDL class diversity with a mix of crops, primarily rice (6.7%) and alfalfa (3.1%), and a diversity of non-crop classes.

3. Methods

3.1. Linear harmonic model

Harmonic analysis using Fourier series, i.e., superimposed sequences of sines and cosines, has been used to model the temporal variation of vegetation using satellite data for several decades (Menenti et al., 1993; Olsson and Eklundh, 1994; Moody and Johnson, 2001). A straightforward definition of the linear harmonic model is:

$$f(t) = a_o + \sum_{m=1}^{M} \left(a_m cos\frac{2\pi t}{L} + b_m sin\frac{2\pi t}{L} \right) \quad (1)$$

where $f(t)$ is the modelled satellite reflectance, or vegetation index value (as in this study), for a single pixel location; a_0 is the Fourier coefficient describing the mean of $f(t)$ over the time series; a_m and b_m are Fourier coefficients for harmonic component m with $M \geq 1$ components; t is time; and L is the length of the time period. In this study $L = 365$, as there are 365 days in 2010, and t is set as the day of the year in 2010 of each Landsat acquisition.

The optimal setting of M is dependent on the temporal surface variation, and so is land cover and condition dependent. If the surface variation is periodic and known a priori, then a suitable value of M can be determined. From classical sampling theory, the satellite observations should be sampled uniformly in time with a frequency that is at least twice the frequency of the surface variations (Shannon, 1949). In practice, Landsat observations are not sampled uniformly in time. For example, although Landsat 5 TM and Landsat 7 ETM+ data are sensed typically 8 days apart, they may be sensed only one day apart where the sensor orbit edges overlap (Roy et al., 2016), and cloud obscuration may occur with distinct seasonality and randomly across individual Landsat images (Kovalskyy and Roy, 2015). Using larger values of M may lead to over fitting that causes spurious oscillations particularly with noisy data and/or near gaps in the time series when surface changes occur (Jakubauskas et al., 2001; Hermance, 2007; Brooks et al., 2012). Given the complexity of these factors, data driven approaches (i.e., trial and error, or cross-validation using independent time series test data) are ideally required to define an appropriate M for each pixel location and time period. Zhu et al. (2015) recommended broadly that for Landsat-based land cover change monitoring, there should be at least three times more valid surface observations than the number of linear harmonic model coefficients, and suggested models with $M = 2$ (i.e., five Fourier coefficients) and $M = 3$ (i.e., seven Fourier coefficients) requiring at least 15 and 21 Landsat observations respectively for the model defined as (1).

3.2. Non-linear harmonic model

Linear harmonic models cannot reliably fit time series with certain types of non-linear variation such as non-symmetric waveforms that cannot be modelled by superposition of sequences of sines and cosines over time. In this study, the non-linear harmonic model adopted by Carrão et al. for MODIS time series fitting (2010) was considered. The model is more complex to implement than the linear model due to the nested trigonometric terms:

$$f(t) = a_o + a_1 \bullet cos\left[\frac{2\pi}{L}t + \phi + \alpha \bullet sin\left(\frac{2\pi}{L}t + \varphi \right) \right] \quad (2)$$

where $f(t)$ is the modelled satellite reflectance, or vegetation index value (as in this study), for a single pixel location; a_0 is equivalent to the

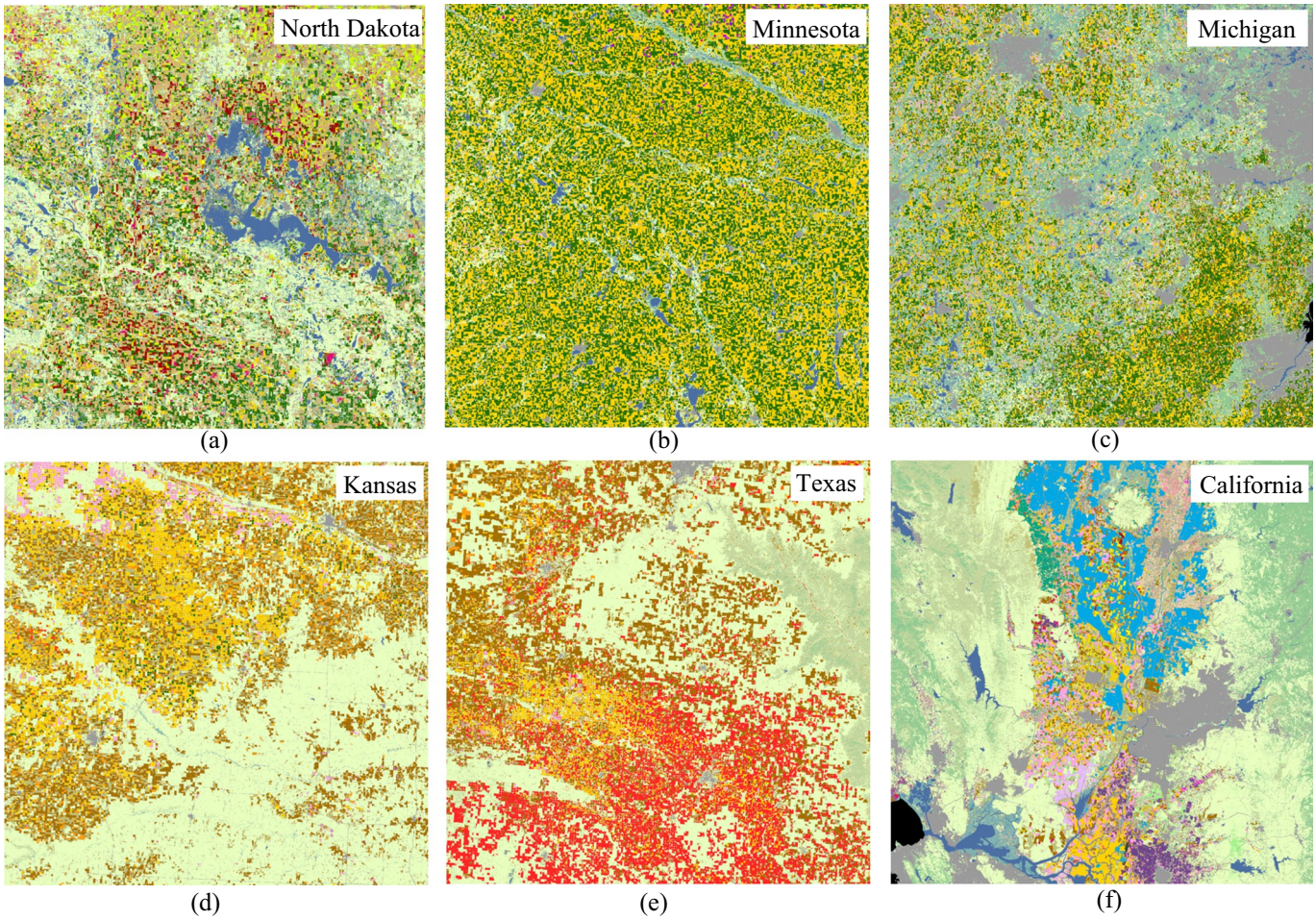


Fig. 3. The 2010 CDL 30 m data for the six ARD tiles shown in Fig. 1 (major classes summarized in Table 1) colored as the CDL web site data (https://nassgeodata.gmu.edu/CropScape/). In the California tile (f), the blue class to the north is rice and the purple class to the south is grapes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

mean of $f(t)$ over the time series; a_1 fixes the peak deviation from the annual mean; ϕ is the annual phase related to the seasonality of the waveform; α controls the degree of non-symmetry of the waveform when $\alpha > 0$; φ is the annual non-linear phase that defines the form of the waveform non-symmetry; t is time and L is the length of the time period that are defined as for (1). The term $\alpha \bullet sin\left(\frac{2\pi}{L}t + \varphi\right)$ provides temporal warping (acceleration/deceleration) of the waveform and enables modelling of intra-annual variations e.g., due to rapid vegetation changes (Carrão et al., 2010). The non-linear model (2) has only five parameters. Despite this small number, the model has been demonstrated to closely fit MODIS time series for a variety of land covers, including agricultural ones, and is able to model double-cropping and triple-cropping and flat temporal trajectories, and is robust to noise and missing observations (Carrão et al., 2010).

3.3. Model implementation and evaluation

The models (1) and (2) were applied to the Landsat atmospherically corrected NDVI time series at each ARD pixel location. For brevity, we refer to the atmospherically corrected NDVI as surface NDVI. The models were solved by least-squares adjustment rather than by analytical expansion of the model terms and then application of linear regression to the expanded terms. The model parameters were initialized with an initial guess and then solved iteratively by least-squares adjustment of the differences between the observed times series values and model-predicted values (Ghilani, 2010). For the linear harmonic model, a_0 was initialized as the median of the surface NDVI time series values, and the other parameters were initialized with zero values. For the non-linear harmonic model, the five parameters were initialized as described in Carrão et al. (2010). The least-squares adjustment, like linear regression, assumes that $f(t)$ has normally distributed measurement errors and that the independent variable (t) is without error. The day of Landsat acquisition, t, is known, but errors in the surface NDVI, $f(t)$, will occur. The NDVI to first order reduces Landsat bi-directional reflectance effects due to solar and view zenith angle changes (Gao et al., 2014; Roy et al., 2016), but is sensitive to residual atmospheric correction errors (Ju et al., 2012). In addition, the ARD pre-processing (Section 2) to remove cloud, shadow and snow pixels, is imperfect. In particular, residual clouds and shadows often constitute significant outliers in satellite time series and are not normally distributed error sources (Jonsson and Eklundh, 2002; Ju et al., 2010). Generally, residual atmospheric correction errors and clouds depress NDVI and so NDVI time series fitting algorithms have been developed to preferentially select the upper envelop of NDVI time series (Jonsson and Eklundh, 2002). However, shadows can have quite variable effects on atmospherically corrected NDVI including increased values (Park, 2013; Zhang et al., 2015). In this study, given the typically sparse and sometime irregular temporal density of Landsat observations, no outlier detection could be undertaken reliably.

The linear and non-linear harmonic models were applied independently at each ARD pixel location where there was at least three times more valid year 2010 Landsat 5 TM and 7 ETM+ NDVI values than the number of model parameters. Valid observations are those that

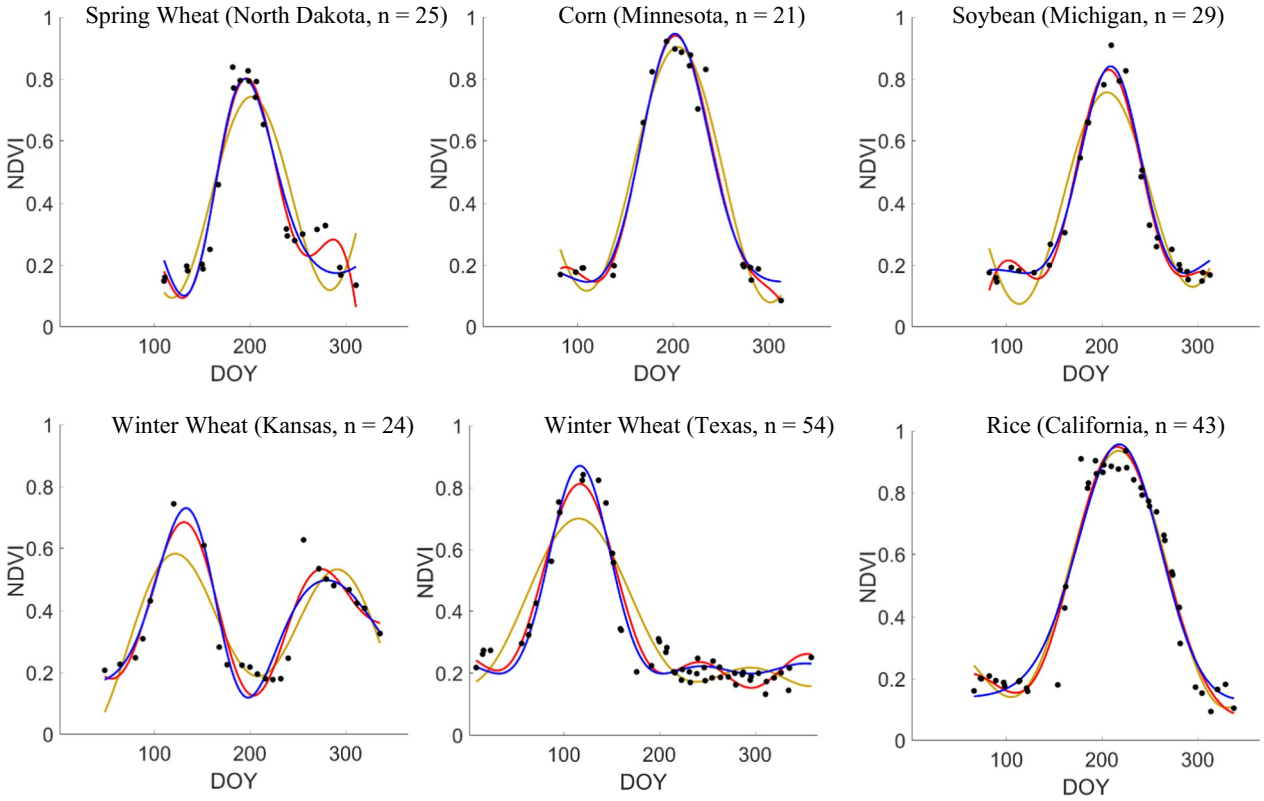


Fig. 4. Example Landsat 5 TM and 7 ETM+ surface NDVI time series for 2010 (black dots) at single ARD pixel locations (Table 2) each extracted from one of the six ARD tiles (Fig. 3). The model fitted results for the linear harmonic model with 5 parameters (yellow) and 7 parameters (red), and for the 5-parameter non-linear harmonic model (blue), are shown. The NDVI data are atmospherically corrected and filtered to remove cloud, snow, shadow, and pixels with surface NDVI < 0.0. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Summary of the six single ARD pixel locations, CDL crop types, and model fitted RMSD (Eq. (3)) values shown in Fig. 4. The model with the lowest RMSD value is shown in bold where LH-5 and LH-7 denote the 5 and 7-parameter linear harmonic models and NLH-5 denotes the 5-parameter non-linear harmonic model.

	North Dakota	Minnesota	Michigan	Kansas	Texas	California
Crop type	Spring Wheat	Corn	Soybean	Winter Wheat	Winter Wheat	Rice
ARD tile pixel location	(1805, 210)	(1809, 213)	(1817, 210)	(1909, 117)	(1909, 209)	(1749, 233)
LH-5 RMSD	0.122	0.067	0.079	0.098	0.083	**0.071**
NLH-5 RMSD	0.075	0.062	**0.043**	0.077	**0.050**	0.077
LH-7 RMSD	**0.072**	**0.058**	0.047	**0.076**	0.057	**0.071**

were unsaturated, had surface NDVI > 0.0, and were not flagged in the ARD as cloud, snow or shadow. The linear model with $M = 2$ (five parameters) and $M = 3$ (seven parameters), and the five-parameter non-linear model, were considered in this study. The model fitting was evaluated quantitatively at each ARD pixel location as:

$$RMSD = \sqrt{\frac{\sum_i^n (NDVI_i - \widehat{NDVI_i})^2}{n - m}} \quad (3)$$

where $NDVI_i$ is the observed and $\widehat{NDVI_i}$ is the modelled Landsat surface NDVI on day i of 2010, n is the number of valid Landsat surface NDVI values at the ARD pixel location, and m is the number of model parameters (five or seven).

4. Results

4.1. Illustrative single pixel time series model fitting examples

Fig. 4 shows example single ARD pixel Landsat NDVI time series (black dots) and the linear and non-linear harmonic model fits (colored lines) for each of the six tiles (Fig. 3). Other pixel locations have different results that are explored for all the tile data in the following sections. The six pixels were selected to be representative of the primary CDL crop class in each tile, namely spring wheat, corn, soybean, winter wheat, and rice (Table 1) and where there were ≥21 annual observations so that the 7-parameter linear model could be implemented reliably. The number of observations varies from 21 (Minnesota) to 54 (Texas). To be objective, the pixels were selected at approximately the same tile locations (see Table 2). In Fig. 4, the sometimes irregular Landsat 5 TM and Landsat 7 ETM+ temporal sampling and the reduced number of winter observations due to snow and cloud is evident. The Texas example is a notable exception because cloud and snow is minimal, which has been observed in previous CONUS Landsat cloud studies (Ju and Roy, 2008; Kovalskyy and Roy, 2015).

The linear and non-linear harmonic models all capture broadly the seasonal evolution of the crops, including the Kansas winter wheat double cropping example that has two distinct NDVI peaks. In general, except for the Kansas example, the 5-parameter (yellow lines) and 7-parameter linear (red lines) linear model fits have noticeable oscillations at the beginning and at the end of the time series. These oscillations are well documented and often considered as a major

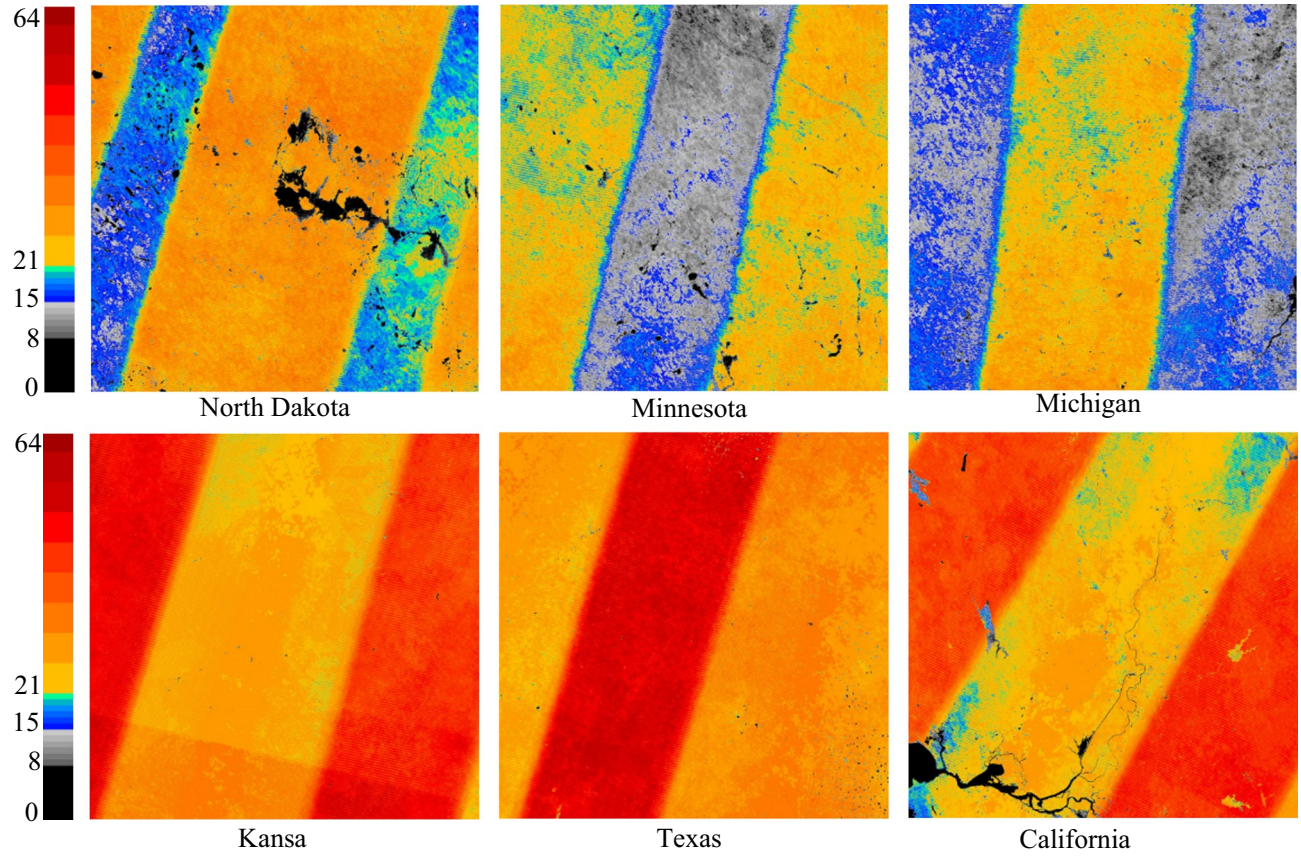


Fig. 5. Total number of annual valid (cloud-free, snow-free, shadow-free, unsaturated, atmospherically corrected NDVI > 0.0) Landsat 5 TM and Landsat 7 ETM+ ARD observations in 2010. Each ARD tile is 5000 × 5000 30 m pixels.

disadvantage of the linear harmonic model (Jakubauskas et al., 2001; Hermance, 2007; Brooks et al., 2012). The 5-parameter linear model (yellow lines) qualitatively appears to perform the least well of the three models; it has both oscillation problems, and the magnitude and/or date of the peak NDVI are not well fitted for any of the illustrated examples except for the California rice example. The 5-parameter non-linear model (blue lines) has fewer of these problems and visually appears to provide the best fits.

Table 2 quantifies the RMSD values for the six examples illustrated in Fig. 4. Except for the California rice example, where the RMSD values for three models are negligibly different (third decimal place in surface NDVI units), the 5-parameter linear harmonic model has the largest RMSD values. The RMSD values for the 7-parameter linear and 5-parameter non-linear models are similar, and they each have the lowest RMSD values for four and two of the six examples respectively. Although the RMSD provides an overall measure of the time series fitting quality, it is not necessarily an appropriate measure for certain applications. For example, the evident oscillations may constitute a significant issue for change detection applications.

4.2. Tile results

4.2.1. Annual number of valid Landsat observations and model RMSD values

Fig. 5 illustrates the total annual number of valid Landsat 5 TM and Landsat 7 ETM+ observations at each ARD pixel location. The number of observations are colored to show where there were < 8 (black), 8 to 14 (gray shades), 15 to 20 (blue and green colors), and ≥ 21 observations (yellow, orange and red colors). The percentage of ARD pixels with ≥15 valid observations varied from 67.31% (Michigan) to 99.97% (Kansas) and the percentage with ≥21 valid observations varied from 39.7% (Michigan) to 99.6% (Texas). The median annual number of valid observations per ARD tile pixel varied among the six tiles: 26 (North Dakota), 21 (Minnesota), 17 (Michigan), 32 (Kansas), 32 (Texas), and 28 (California). These numbers vary due to snow and cloud contamination, which is particularly apparent in the northern ARD tiles (Figs. 2 and 5, i.e., North Dakota, Minnesota, Michigan), and due to the Landsat orbit and sensing geometry. The edges of the Landsat 5 TM and Landsat 7 ETM+ orbits overlap in the along scan direction (Roy et al., 2016), and so certain geographic locations have more observations per year than others regardless of any snow or cloud contamination. Consequently, the inclination of the Landsat orbit relative to the Albers ARD projection is evident in the spatial distribution of the number of valid observations. Water bodies with < 8 observations are also apparent, for example, in the California and North Dakota tiles, and occur because they frequently have surface NDVI < 0 and were removed from the analysis. The 2003 Landsat 7 ETM+ scan line corrector failure reduced the availability of useful ETM+ data by 22% in strips of missing pixels oriented along-scan (Markham et al., 2004) and this is evident where the annual number of valid observation results change between pixels in the Landsat along-track direction.

Figs. 6 and 7 show the RMSD values for the three model fits (rows) run at each ARD pixel location for the six ARD tiles. Only the RMSD values where there were ≥ 8 annual valid observations are shown. The RMSD values vary among the six tiles reflecting the availability of valid observations (Figs. 2 and 5) and the different land covers at each ARD pixel location (Fig. 3). In general, the RMSD values for the 5-parameter linear harmonic model (top row) are greater than the 5-parameter non-linear harmonic model (middle row) and 7-parameter linear harmonic model (bottom row) values. The models (1) and (2) can only be resolved if there are at least as many observations as model coefficients and the RMSD is only meaningful when there are a much larger number

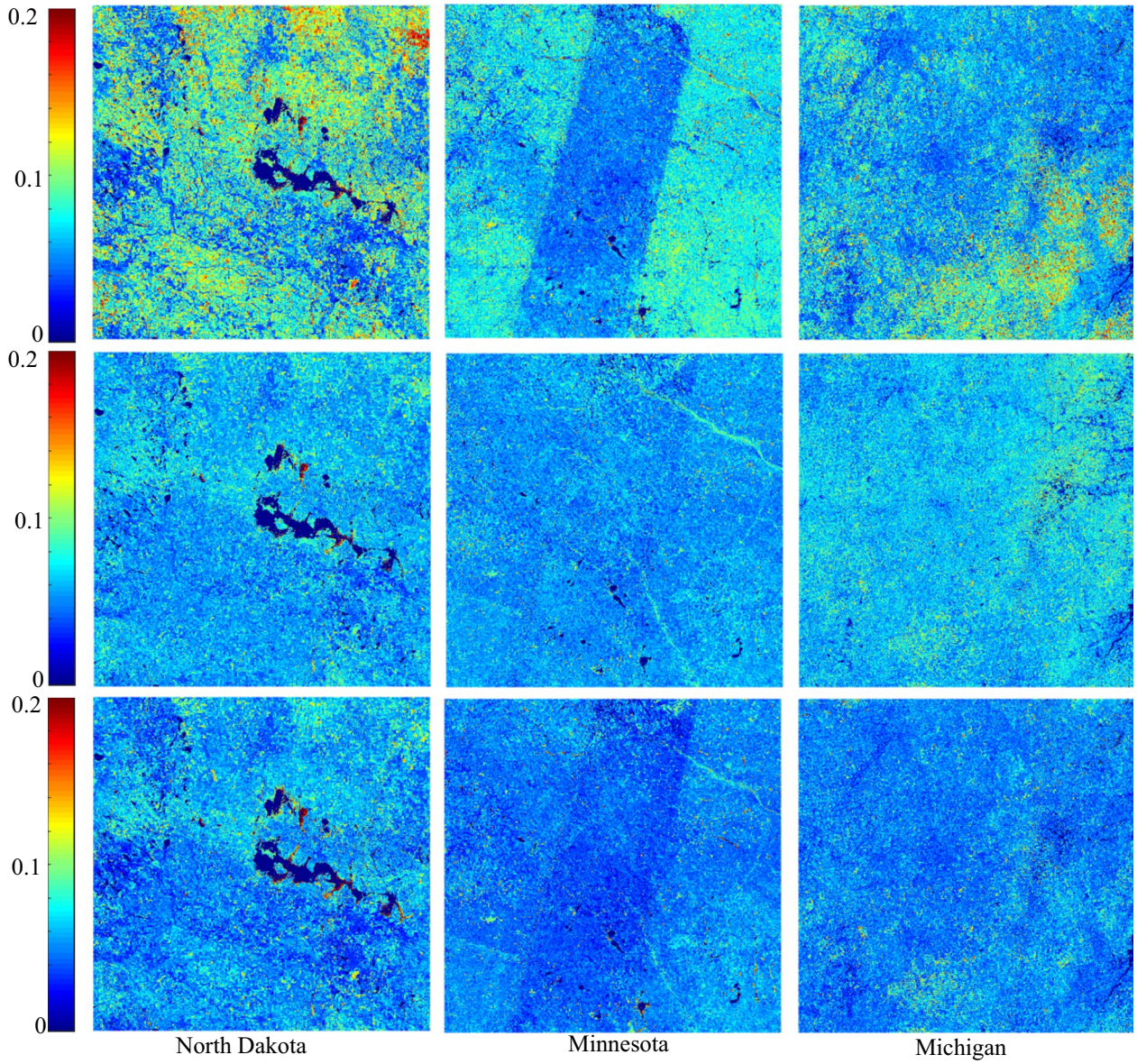


Fig. 6. RMSD fitting values (Eq. (3)) for the 5-parameter linear (top row), 5-parameter non-linear (middle row) and 7-parameter linear (bottom row) harmonic models fit to every ARD 30 m pixel location with ≥8 annual valid observations in the North Dakota, Minnesota and Michigan tiles.

of observations. Following the recommendation that there should be at least three times more observations than the number of model coefficients (Zhu et al., 2015), the 5 and 7-parameter model RMSD values are subsequently only considered where there are at least 15 and 21 observations respectively. Thus, the illustrated RMSD values in the broad swaths of the northern ARD tiles (Fig. 6, North Dakota, Minnesota, Michigan) with 8–15 or 8–20 annual observations have unreliable RMSD values for the 5 and 7-parameter models, respectively, and are not considered in the rest of this study for these models. In the more southern ARD tiles (Fig. 7), Landsat data availability is less of an issue, except for a portion of the California tile where there were fewer than 21 annual observations.

4.2.2. Crop and non-crop analysis

Land-cover-specific, i.e. CDL-class-specific, interpretation of the RMSD values shown in Figs. 6 and 7 is complex because of the large number of classes, their different areal coverages, the different availability and timing of the Landsat observations, and because the same CDL class may have quite different seasonality among the six tiles due to edaphic, weather, crop planting date and crop treatment factors. For example, the Kansas and Texas winter wheat surface NDVI time series in Fig. 4 have quite different seasonality but the same CDL class. However, a pattern emerges when the RMSD values are analyzed in terms of crop and non-crop land covers.

Fig. 8 shows crop and non-crop histograms of the 5-parameter non-linear and linear model RMSD values derived where there were ≥15 annual valid observations. Table 3 provides summary statistics of the RMSD values. Only results for the 5-parameter models are considered here as the 7-parameter model requires ≥21 annual valid observations. For all six tiles, the crop RMSD values (solid lines) are smaller and have narrower histograms for the 5-parameter non-linear model (blue) than for the 5-parameter linear model (yellow). Among the six tiles, the California tile has the greatest mean and standard deviation for the 5-parameter non-linear model crop RMSD values. This is due to the considerable diversity of crop types (Table 1) including crops that are not reliably fit by the models (examples are shown later). When the non-crop classes (Fig. 8, dashed lines) are considered, the difference between the two models is less clear, although in general, the non-linear model provides smaller mean and standard deviation RMSD values than the linear model. However, the Michigan tile has a smaller

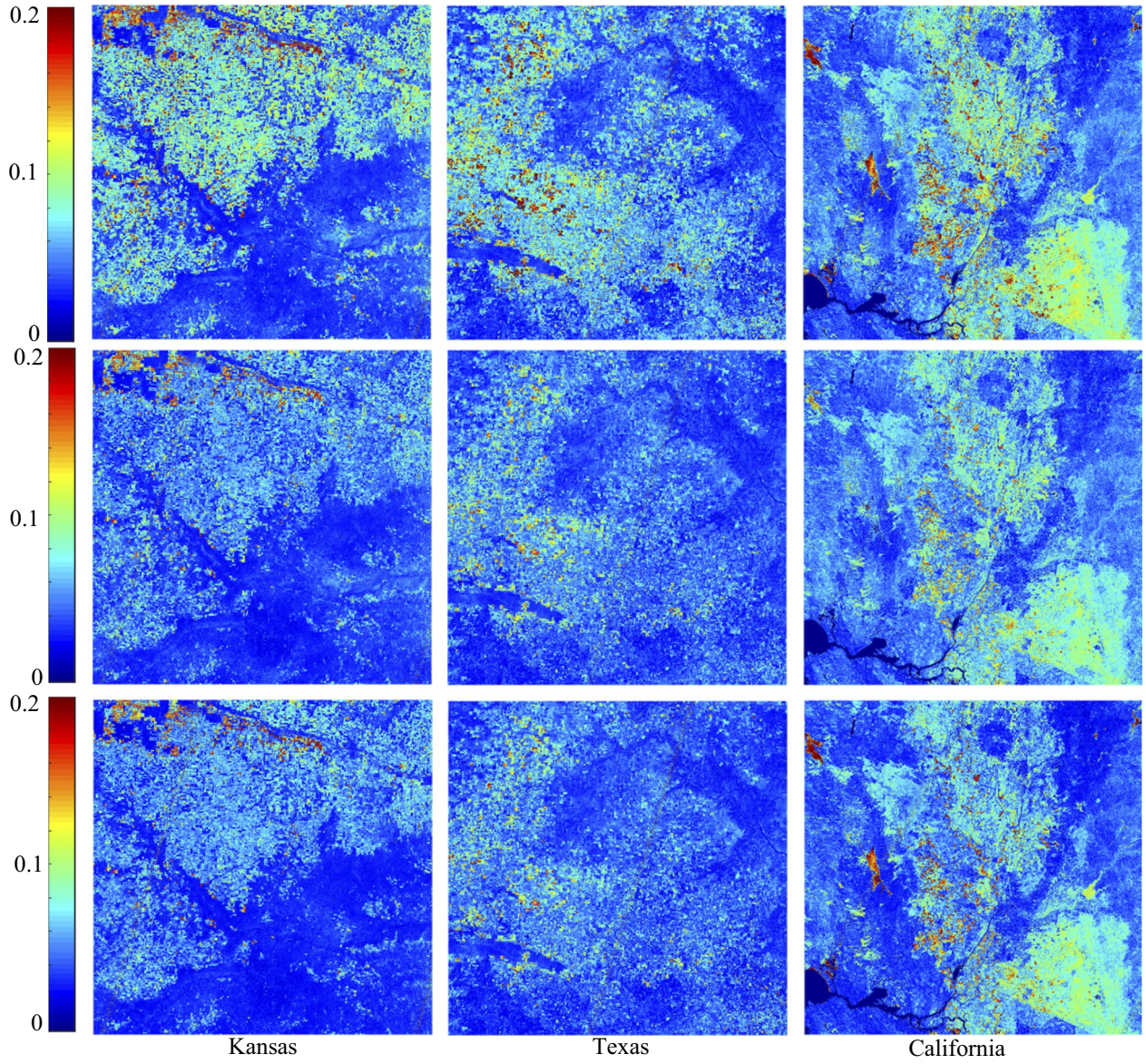


Fig. 7. RMSD fitting values (Eq. (3)) for the 5-parameter linear (top row), 5-parameter non-linear (middle row) and 7-parameter linear (bottom row) harmonic models fit to every ARD 30 m pixel location with ≥8 annual valid observations in the Kansas, Texas and California tiles.

mean RMSD for the 5-parameter linear model, and careful examination indicates that this is because the 5-parameter non-linear model is often less able to reliably model the predominant (Table 1) deciduous forest class (examples are shown later).

Fig. 9 shows the crop histograms for the three models where there were ≥21 annual valid observations and the RMSD values are summarized in Table 4. Results for the three models (i.e., for the 7-parameter linear model and both the five parameter models) are shown as ≥21 observations are sufficient to fit all the models. The differences between the 5-parameter non-linear and the 5-parameter linear model crop RMSD values observed above in Fig. 8 are also observed with ≥21 valid observations. The 5-parameter linear model again performs poorly over crops - for all the tiles, the RMSD values are greater with wider histograms than the other models. The 7-parameter linear and 5-parameter non-linear model crop RMSD values are similar. However, for all the tiles, the 7-parameter linear model has marginally smaller mean RMSD (in the third decimal place in surface NDVI units) and larger (or the same for one tile) RMSD standard deviation values.

Among the six tiles, the California tile has the greatest mean and standard deviation crop RMSD values for all three models (Fig. 9, Table 4). This is due to the diversity of crops (Table 1) including crops that are not reliably fit, which is illustrated in Fig. 10 for alfalfa and almond single pixel time series examples. For the alfalfa examples, it appears that the ARD filtering (Section 2) did not perform correctly and so the NDVI data are temporally noisy. Although the filtering is imperfect, the temporal variation in NDVI is also due to multiple alfalfa cropping cycles under irrigated conditions, which has been observed for alfalfa elsewhere (Houborg and McCabe, 2018), and in this example cannot be captured reliably by only using Landsat 5 TM and Landsat 7 ETM+ temporal resolution data. The almond example exhibits two NDVI peaks of similar magnitude several months apart, and all the models provide poor fits – both the 5-parameter models fail to capture the reduced NDVI between the peaks, and the 7-parameter linear model captures it but the NDVI oscillates elsewhere. The California non-crop grass/pasture example is included because many of the high RMSD values at the South-East corner of the California tile (Fig. 7) occur over this CDL class (Fig. 2, yellow-green), and as illustrated in Fig. 10 are sometimes not well fit by any model.

Fig. 11 shows the non-crop histograms for the three models where there were ≥21 annual valid observations (summarized in Table 5).

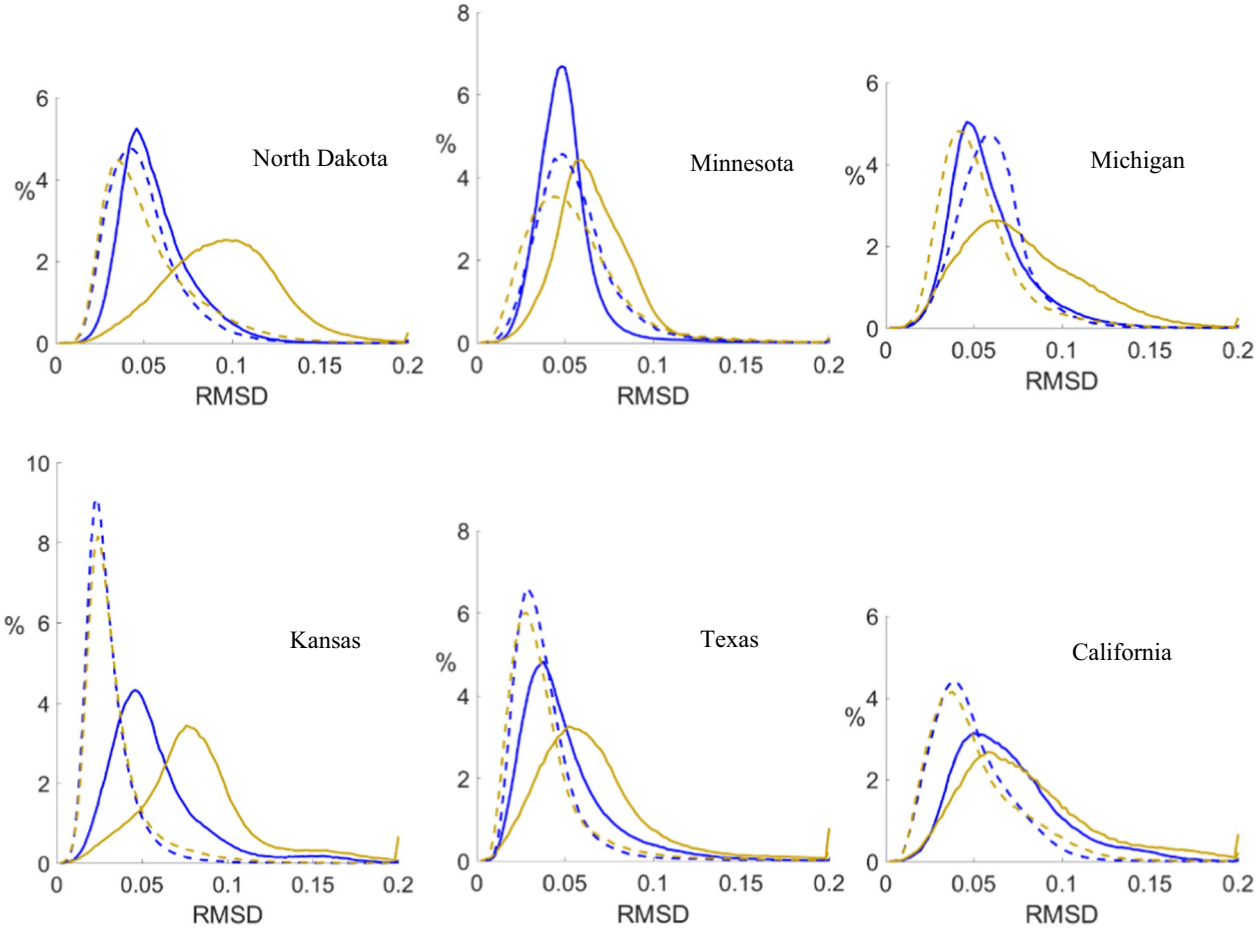


Fig. 8. Histograms of the 5-parameter non-linear (blue) and 5-parameter linear (yellow) RMSD values (Figs. 6 and 7) derived for the pixels that are labelled by the CDL (Fig. 2) as crop (solid lines) or non-crop (dashed lines) and have ≥15 valid annual Landsat observations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
RMSD summary statistics (mean μ and standard deviation σ) of the Fig. 8 results, i.e., of the 5-parameter linear (LH-5) and non-linear (NLH-5) model fits for the crop and non-crop pixels, derived only where there were ≥15 valid annual Landsat observations. The percentages summarize the percentage of ARD tile pixels (from a total of 5000^2 pixels). The model result with the lowest mean RMSD value is shown in bold for each ARD tile.

	North Dakota		Minnesota		Michigan		Kansas		Texas		California	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
Crop pixels												
LH-5	0.100	0.030	0.075	0.021	0.083	0.032	0.089	0.032	0.072	0.032	0.085	0.036
NLH-5	**0.064**	0.019	**0.055**	0.016	**0.064**	0.021	**0.065**	0.027	**0.054**	0.024	**0.074**	0.029
%	48.9%		58.3%		30.9%		35.3%		43.9%		23.6%	
Non-crop pixels												
LH-5	0.058	0.026	**0.061**	0.026	**0.057**	0.022	0.041	0.019	0.045	0.024	0.056	0.027
NLH-5	**0.056**	0.019	**0.061**	0.021	0.065	0.018	**0.037**	0.014	**0.042**	0.017	**0.053**	0.021
%	43.9%		14.4%		36.4%		64.7%		55.9%		73.7%	

The 7-parameter linear model has the smallest mean non-crop RMSD value for all the tiles, and the 5-parameter non-linear model has the smallest standard deviation for all the tiles. The smaller non-linear model RMSD standard deviation was also observed for the crop pixels (Table 4). This is likely due to the relative robustness of the non-linear model to over-fitting and oscillations compared to the linear models. This is evident in the single pixel time series examples shown in Figs. 4, 10 and 12.

The 5-parameter linear model performs relatively poorly for the non-crop pixels and has the greatest mean and standard deviation RMSD for all the tiles (Table 5) except for the Michigan tile where the 5-parameter non-linear model has a greater mean RMSD value. The predominant non-crop class in the Michigan tile is deciduous forest (13.8% of the tile, Table 1) and the phenology of this class is usually not well fitted by any of the models. This is illustrated in Fig. 12 for two Michigan deciduous forest single pixel time series examples. Difficulties in harmonic linear model fitting of the characteristic deciduous forest NDVI time series dome shape have been noted as a reason for the development of logistic regression curves for MODIS phenology studies over these kinds of land cover (Zhang et al., 2003). Fig. 12 also shows a Michigan alfalfa example, that is reasonably well fitted by the 5-parameter non-linear model.

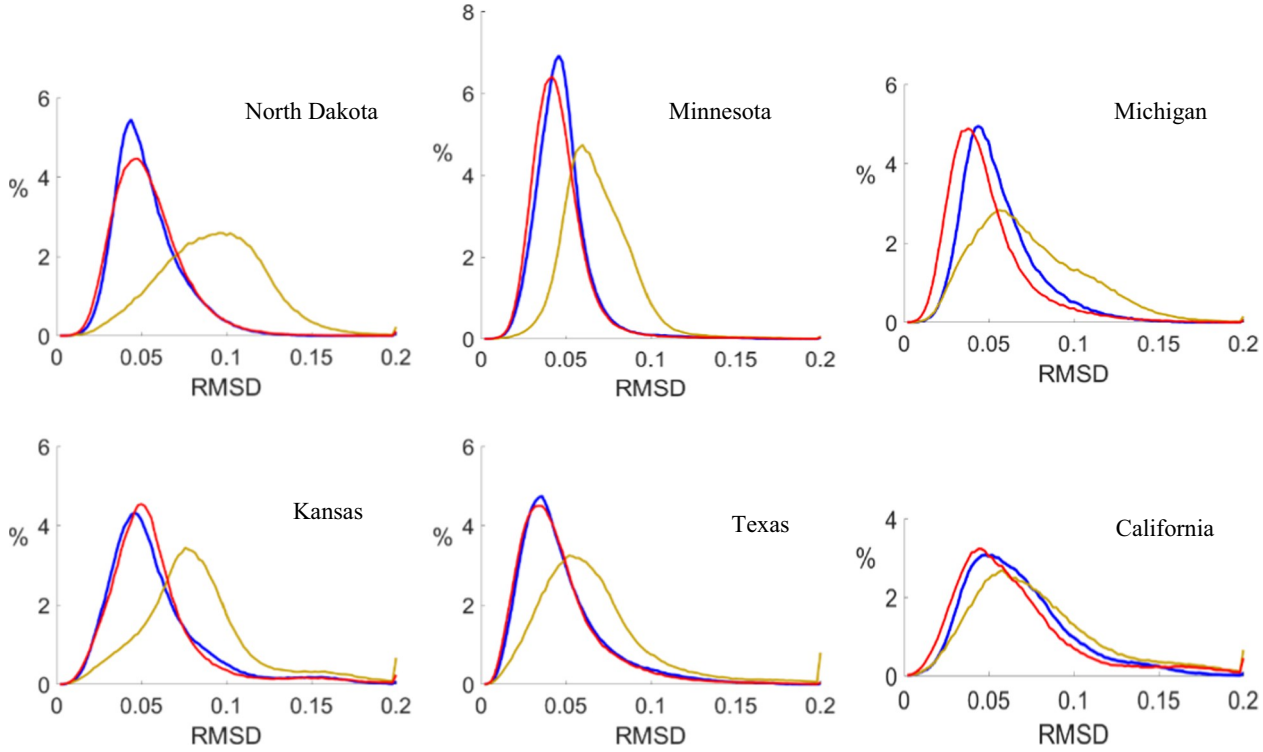


Fig. 9. Crop pixel histograms of the 5-parameter non-linear (blue), 5-parameter linear (yellow) and 7-parameter linear (red) RMSD values (Figs. 6 and 7) derived for CDL crop pixels where there were ≥ 21 valid annual Landsat observations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4
Crop pixel RMSD summary statistics of the Fig. 8 results, i.e., of the 5-parameter linear (LH-5), 5-parameter non-linear (NLH-5), and 7-parameter linear (LH-7) model fits, derived only where there were ≥21 valid annual Landsat observations at each CDL crop tile pixel location. The percentages summarize the percentage of ARD tile pixels (from a total of 5000^2 pixels). The model result with the lowest mean RMSD value is shown in bold for each ARD tile.

	North Dakota		Minnesota		Michigan		Kansas		Texas		California	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
LH-5	0.099	0.028	0.077	0.019	0.080	0.031	0.089	0.032	0.072	0.032	0.085	0.036
NLH-5	0.063	0.018	0.055	0.015	0.064	0.021	0.065	0.027	0.054	0.024	0.074	0.029
LH-7	**0.062**	0.020	**0.052**	0.016	**0.057**	0.022	**0.064**	0.028	**0.052**	0.024	**0.072**	0.034
%	37.5%		44.1%		16.8%		35.1%		43.8%		22.9%	

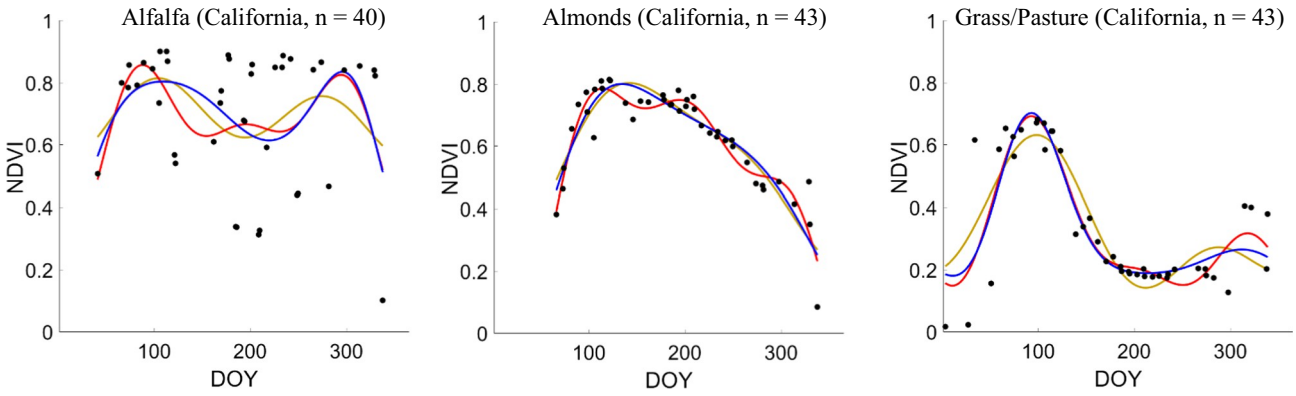


Fig. 10. Example 2010 Landsat 5 TM and 7 ETM+ surface NDVI time series (black dots) at single ARD pixel locations over the California tile. The model fitted results for the linear harmonic model with 5 parameters (yellow) and 7 parameters (red), and for the 5-parameter non-linear harmonic model (blue), are shown. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.2.3. Crop peak NDVI date and magnitude estimation

There is considerable ongoing research on the use of satellite time series for crop status and yield estimation and forecasting. There are a number of approaches, ranging from purely statistical ones using satellite time series, to approaches that use process models with satellite time series and agro-climatic data to parametrize or adjust the process

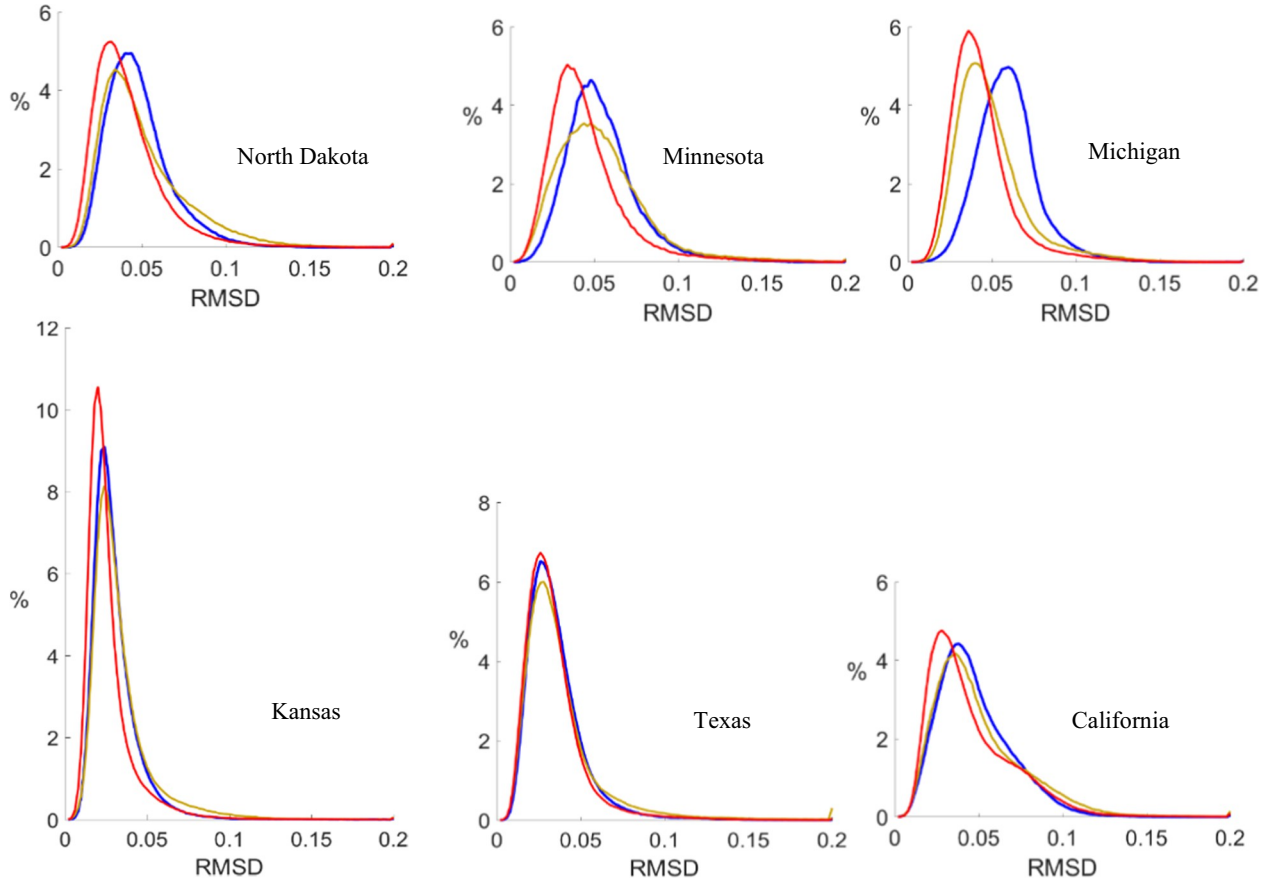


Fig. 11. Non-crop pixel histograms of the 5-parameter non-linear (blue), 5-parameter linear (yellow) and 7-parameter linear (red) RMSD values (Figs. 4 and 5) derived for CDL crop pixels where there were ≥21 valid annual Landsat observations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Non-crop pixel RMSD summary statistics of the Fig. 11 results, i.e., for the 5-parameter linear (LH-5), 5-parameter non-linear (NLH-5), and 7-parameter linear (LH-7) model fits derived only where there were ≥21 valid annual Landsat observations at each CDL non-crop tile pixel location. The percentages summarize the percentage of ARD tile pixels (from a total of 5000^2 pixels). The model result with the lowest mean RMSD value is shown in bold for each ARD tile.

	North Dakota		Minnesota		Michigan		Kansas		Texas		California	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
LH-5	0.058	0.024	0.062	0.025	0.056	0.020	0.041	0.019	0.045	0.024	0.056	0.026
NLH-5	0.055	0.019	0.062	0.020	0.065	0.017	0.037	0.014	0.042	0.017	0.054	0.021
LH-7	**0.049**	0.020	**0.051**	0.021	**0.050**	0.018	**0.034**	0.015	**0.041**	0.018	**0.050**	0.024
%	28.9%		10.6%		23.0%		64.5%		55.8%		69.6%	

model. Many approaches use the date and the amplitude of the maximum (i.e., peak) growing season NDVI that is found by temporal interpolation or model fitting of satellite NDVI time series (Bolton and Friedl, 2013; Franch et al., 2015; Chipanshi et al., 2015; Johnson, 2016; Battude et al., 2016; Azzari et al., 2017).

The harmonic-model time series fitting examples in Fig. 4 indicate variable capability to define the magnitude and date of the peak NDVI over crops. In particular, as noted previously, the 5-parameter linear harmonic model is not reliable for this purpose, and the 5-parameter non-linear and 7-parameter linear harmonic models often define similar peak NDVI dates and amplitudes. To quantify this, the day and magnitude of peak NDVI were derived with the 5-parameter non-linear and 7-parameter linear harmonic models for the crop pixels where there were ≥21 valid Landsat observations in 2010. To resolve ambiguities where crops may have multiple NDVI peaks (e.g., the Fig. 4, Kansas winter wheat example), only the first peak was resolved.

Fig. 13 shows histograms of the day of the year where the peak surface NDVI was predicted by the 5-parameter non-linear (blue) and 7-parameter linear (red) harmonic models for each tile. As expected, the primary crops in each tile (Table 1) dominate the histogram. For example, in the North Dakota tile the two main crops are spring wheat (20.3% of the tile) and soybean (13.5%) and they have distinct surface NDVI peaks at approximately DOY 188 and 221 respectively. The California tile has a greater range of days due to the diversity of crop types and crop calendars. There are evident small differences in the DOY of peak NDVI between the two models. Table 6 summarizes the mean differences derived by pixel comparison of the two model results. The 7-parameter linear harmonic model, on average, estimates the DOY of peak surface NDVI by less than a day to a few days before the 5-parameter non-linear harmonic model. When the absolute differences are considered the mean peak surface NDVI DOY is different between the models on average from < 2 days (North Dakota and Minnesota) to nearly 10 days (California). This is likely because of the shorter growing season and smaller diversity of crop types in North Dakota and

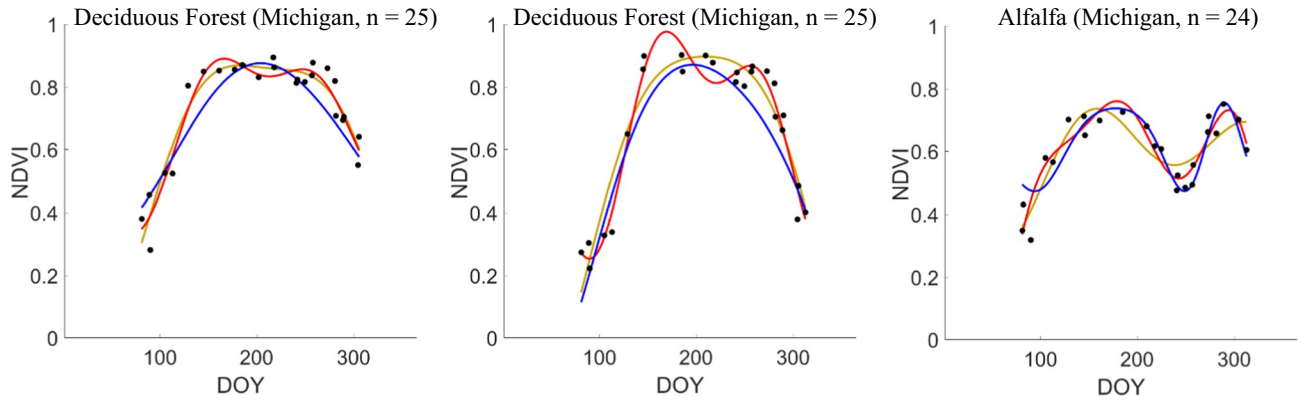


Fig. 12. Example 2010 Landsat 5 TM and 7 ETM+ surface NDVI time series (black dots) at single ARD pixel locations over the Michigan tile. The model fitted results for the linear harmonic model with 5 parameters (yellow) and 7 parameters (red), and for the 5-parameter non-linear harmonic model (blue), are shown. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

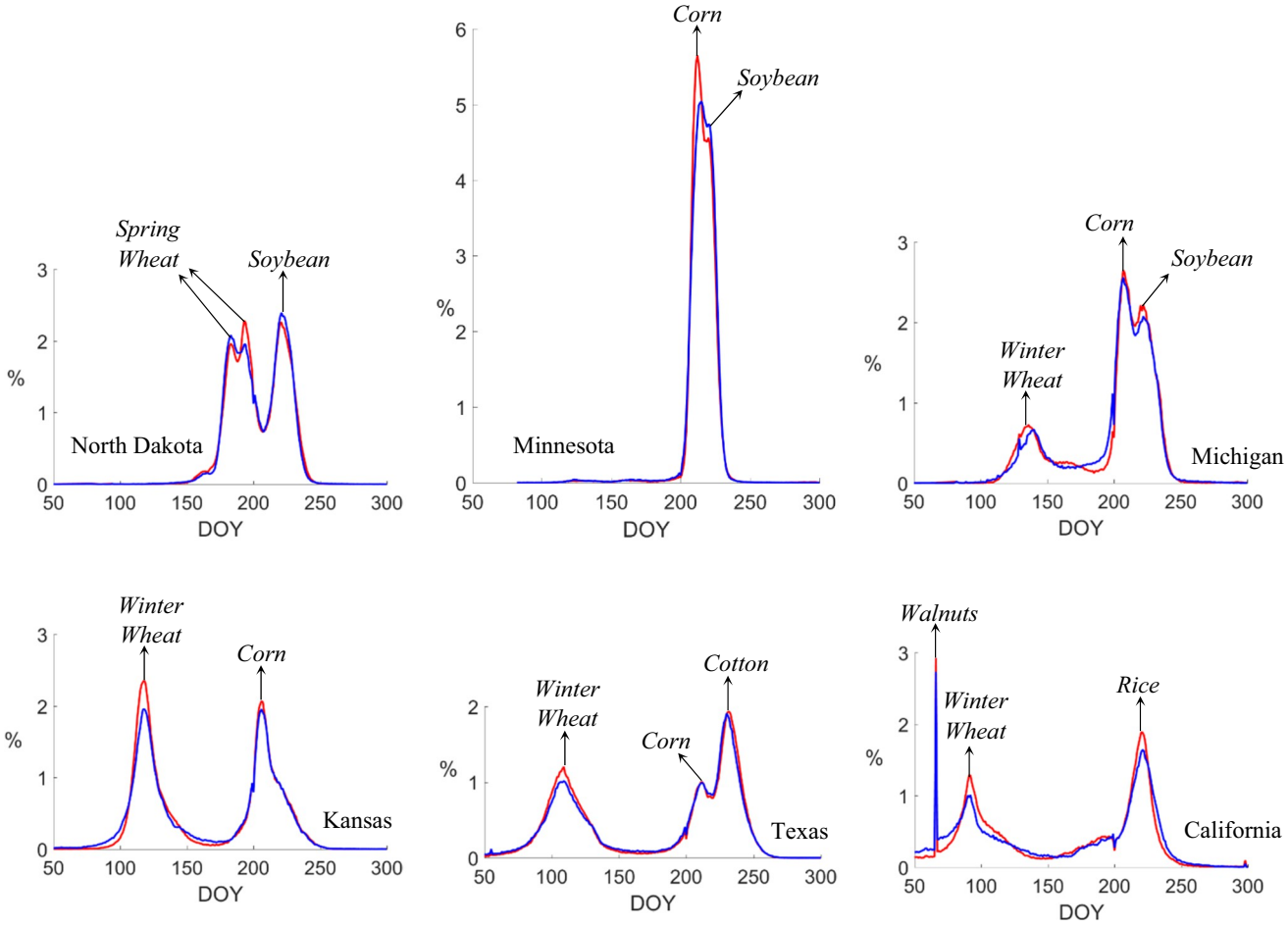


Fig. 13. Histograms of the day of the year (DOY) where the peak surface NDVI was predicted by the 5-parameter non-linear (blue) and 7-parameter linear (red) harmonic models for each tile. Results derived for CDL crop pixels where there were ≥21 valid annual Landsat observations. The arrows and CDL class name annotations denote the majority CDL class that had the corresponding DOY NDVI peak values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Minnesota compared to California which also includes crops that are less easily modelled (Fig. 10). Comparison of the Fig. 13 results with Fig. 2, which shows the spatial percentage of ARD tile pixels with valid observations in 2010, indicates that there is no correspondence between the peak surface NDVI dates and the growing season temporal data availability.

Histograms of the magnitude of peak surface NDVI predicted by the two models for each tile are not shown as they are not particularly informative (they reflect primarily differences in the peak NDVI of the different crops in each tile). As observed in Fig. 4, the 5-parameter non-linear harmonic model peak surface NDVI tends to be slightly greater than the peak value predicted by the 7-parameter linear harmonic model. On average this is the case for four of the six tiles but with negligible difference on average between the models for the Michigan

Table 6
Summary statistics of Δ_{peak_doy} = (DOY of NDVI peak for 7-parameter linear harmonic model) – (DOY of NDVI peak for 5-parameter non-linear harmonic model) derived at each crop pixel for each tile. Results derived for crop pixels where there were ≥21 valid annual Landsat observations. The percentages summarize the percentage of ARD tile pixels (from a total of 5000^2 pixels).

	North Dakota	Minnesota	Michigan	Kansas	Texas	California
$\mu(\Delta_{peak_doy})$	−0.1	−0.5	−2.2	−1.7	−0.9	−3.6
$\sigma(\Delta_{peak_doy})$	5.0	4.5	15.1	16.1	15.1	20.6
$\mu(\lvert\Delta_{peak_doy}\rvert)$	1.9	1.8	6.6	5.1	4.8	9.6
%	37.5%	44.1%	16.8%	35.1%	43.8%	22.9%

and California tiles (Table 7). When the absolute differences are considered, the mean peak NDVI is different between the models on average by < 0.03 (NDVI units) for all the tiles expect for the Kansas tile (0.042).

5. Discussion and conclusion

The primary finding of this study is that the 7-parameter linear harmonic model and the 5-parameter non-linear harmonic model (requiring at least 21 and 15 observations respectively) can both be used to fit the annual temporal variation of CONUS crops. The use of 30 m Landsat data, rather than coarser spatial resolution data such as MODIS, is attractive due to the ability to capture within-field variations (edaphic, crop treatment, crop disease, drainage etc.) and to reduce ambiguities in regions where fields are small and/or spatially fragmented (White and Roy, 2015; Fritz et al., 2015). Two Landsat sensors are required, however, to provide sufficient annual surface observations for the model fitting. Crops with two maturity stages, such as winter wheat can be modelled, but crops with multiple stages, such as alfalfa, are less reliably modelled due to their temporal complexity that is insufficiently sampled even using combined Landsat 5 TM and Landsat 7 ETM+ time series data.

Landsat 5 TM and Landsat 7 ETM+ provide an 8-day revisit i.e., about 45 observations per year, but cloud and snow contamination and the Landsat orbit and sensing geometry reduce the number of valid observations considerably. For example, in the six Landsat 150 × 150 km ARD tiles considered, the median annual number of valid observations per ARD tile pixel varied from 17 (Michigan) to 32 (Kansa and Texas). For example, the percentage of pixels with ≥15 valid observations in 2010 varied from about 67% (Michigan) to 100% (Kansas), and the percentage with ≥21 valid observations varied from about 40% (Michigan) to 100% (Texas). Thus, for the Michigan tile, even if a 5-parameter model is used, 33% of the pixels cannot be modelled reliably, and if the 7-parameter linear model is used, 60% cannot be modelled reliably. At locations where there are insufficient observations, the models are unreliable and should not be used. A potential solution is to use medium resolution Sentinel 2A, Sentinel 2B and Landsat 8 time series data to improve the temporal density of observations. The combination of these sensor data will provide a global median average revisit interval of 2.9 days, and so will provide weekly cloud-free observations assuming 50% cloud cover at the time of satellite overpass (Li and Roy, 2017). However, sensor combination is non-trivial, and several challenges remain in processing these different sensor data so that they can be used together reliably (Storey et al., 2016; Yan et al., 2018; Zhang et al., 2018). If the incidence of missing or unreliable Landsat observations is not spatially correlated locally, and if within-field variations and field small axis dimensions are greater than several Landsat pixels, then time series fitting models could be applied to locally spatially aggregated data (e.g., to medium values derived over 3 × 3 pixel windows). This could help reduce the requirement for sufficient temporal observations needed for the model fitting.

If there are < 21 annual Landsat observations, the 5-parameter *non-linear* harmonic model is recommended for fitting annual NDVI time series. This is because the 5-parameter linear harmonic model performed unambiguously poorly. It had greater tile RMSD values over crop and non-crop pixels compared with the 5-parameter non-linear model for pixels with ≥15 valid annual observations, and also compared with the 7-parameter linear and the 5-parameter non-linear models for pixels with ≥21 observations. In the illustrated single pixel crop and non-crop time series examples, the 5-parameter linear harmonic model had evident oscillation fitting problems and the magnitude and date of the peak NDVI was evidently not as well fitted as the other models.

If there are ≥21 annual Landsat observations, the results of this study indicate that either the 5-parameter non-linear or the 7-parameter linear model can be used to fit annual crop NDVI time series. Regarding the statistics of model fitting RMSD, the 7-parameter model had marginally smaller mean values, but larger standard deviations than the 5-parameter non-linear model for the six ARD tiles. The smaller RMSD standard deviation values are likely due to the relative robustness of the non-linear model to over-fitting and oscillations compared with the 7-parameter linear model. However, there is no way to validate which model fit is more accurate without sufficient and representative independent reference data, which are unavailable.

The RMSD was used as an overall measure of the model time series fitting quality. It is not necessarily an appropriate measure for certain applications. This paper focused on the ability of the models to fit annual Landsat NDVI time series, and not change detection or early growing season NDVI forecasting that is of interest for the considerable economic importance of this information for crop yield forecasting (Zhong et al., 2016; Gao et al., 2017; Azzari et al., 2017). The predictive capability of the harmonic models to forecast the timing of NDVI peaks, which has been undertaken with MODIS data for example (Franch et al., 2015), was not examined in this study. Research to investigate this is recommended, although the harmonic models do not provide fitting error estimates, which will reduce their utility for forecasting and also change detection applications. Further, the frequent scarcity of Landsat observations in the period between vegetation spring green-up and maturity (e.g., see Fig. 4 examples) suggests that reliable forecasting at Landsat 30 m pixel scale may be challenging without incorporation of some other information or development of new techniques, such as the ones discussed above, or large-area Landsat gap-filling (Yan and Roy, 2018).

The reported analysis of the harmonic models' ability to fit the crop peak NDVI date and peak NDVI value provided insights into their differences, but not on which model was better for this task. The

Table 7
Summary statistics of Δ_{peak_NDVI} = (peak NDVI for 7-parameter linear harmonic model) – (peak NDVI for 5-parameter non-linear harmonic model) derived at each crop pixel for each tile. Results derived for crop pixels where there were ≥21 valid annual Landsat observations. The percentages summarize the percentage of ARD tile pixels (from a total of 5000^2 pixels).

	North Dakota	Minnesota	Michigan	Kansas	Texas	California
$\mu(\Delta_{peak_NDVI})$	−0.017	−0.003	0.002	−0.017	−0.010	0.002
$\mu(\lvert\Delta_{peak_NDVI}\rvert)$	0.025	0.010	0.025	0.042	0.028	0.027
%	37.5%	44.1%	16.8%	35.1%	43.8%	22.9%

differences between the 7-parameter linear and the 5-parameter non-linear harmonic models were not large. The mean absolute differences for each ARD tile in the estimated peak NDVI values was < 0.03 (NDVI units) for all the tiles expect for the Kansas tile (0.042). These differences are small relative to the peak NDVI values (typically 0.8 to 0.9). For each ARD tile, the mean absolute differences in the estimated peak NDVI days varied from < 2 days in the northern tiles, which had short growing seasons and similar crops (North Dakota and Minnesota), to less than a week for the other tiles except for nearly 10 days for the California tile. A 10 day difference in the date of peak NDVI is quite large and is indicative of the noted presence of crops in the California tile, such as alfalfa, that were less easily modelled. The harmonic models' ability to assess other phenological indicators, such as the start or end of the growing season, was not considered in this paper. However, even with a year of combined Landsat 5 TM and Landsat 7 ETM+ data, inspection of select single-pixel time series examples (Figs. 4, 10 and 12) suggest unreliable extraction performance for these phenological indicators.

The focus of this study was on agricultural crops; the six ARD tiles were selected to be representative of the major CONUS crop types. Therefore, the non-crop model fitting analysis results described in this study are not necessarily representative of CONUS non-crop land covers. Notably, however, the Michigan deciduous forest and California grass/pasture time series examples suggest that the harmonic models are less accurate for fitting annual NDVI time series of these land cover types. Further work to investigate this is recommended. The results were reported for 2010 and so may not be the same in other years at the same locations, for example, due to drought or frost that may change the crop phenology.

NDVI time series were used in this study to reduce interpretation complexity and because the NDVI is commonly used for agricultural studies (Johnson, 2016). Other vegetation indices or individual spectral reflectance bands may provide different results. In particular, research to investigate the use of Landsat surface reflectance time series, or nadir bidirectional reflectance distribution function (BRDF) adjusted surface reflectance (Roy et al., 2016), is merited. This is because crops may have quite different spectral reflectance but similar NDVI. For example, in the United States, sunflower and soybeans have different Landsat spectral reflectance but similar NDVI temporal profiles (Yan and Roy, 2015). Regardless of the spectral bands or vegetation indices used, the results of this study demonstrate the need for higher temporal resolution Landsat-like data that has also been advocated for many other terrestrial applications (Roy et al., 2014b).

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